



A review of efficiency criteria suitable for evaluating low-flow simulations

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SUMMARY

Low flows are seasonal phenomena and an integral component of the flow regime of any river. Because of increased competition between water uses, the demand for forecasts of low-flow periods is rising. But how low-flow predictions should be evaluated? This article focuses on the criteria able to evaluate the efficiency of hydrological models in simulating low flows. Indeed, a variety of criteria have been proposed, but their suitability for the evaluation of low-flow simulations has not been systematically assessed.

Here a range of efficiency criteria advised for low flows is analysed. The analysis mainly concentrates on criteria computed on continuous simulations that include all model errors. The criteria were evaluated using two rainfall–runoff models and a set of 940 catchments located throughout France. In order to evaluate the capacity of each criterion to discriminate low-flow errors specifically, we looked for the part of the hydrograph that carries most of the weight in the criterion computation.

Contrary to what was expected, our analysis revealed that, in most of the existing criteria advised for low flows, high flows still make a significant contribution to the criterion's value. We therefore recommend using the Nash–Sutcliffe efficiency criterion calculated on inverse flow values, a valuable alternative to the classically used criteria, in that on average it allows focusing on the lowest 20% of flows over the study period.

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1. Introduction

1.1. In the jungle of efficiency criteria

Hydrological modelling aims at understanding and interpreting catchment hydrological behaviour. It is also used to address a number of practical issues, ranging from flood estimation to water resources management and low-flow forecasting. Whatever model is applied, the model user needs appropriate and meaningful indicators informing on the actual capacity of the model.

However, the evaluation of goodness-of-fit is not as straightforward as it may seem at first glance. Of course, model performance can first be evaluated by the visual comparison of the observed and simulated flow hydrographs, but this remains extremely dependent on the evaluator's experience (Chiew and McMahon, 1993; Houghton-Carr, 1999). A more objective way to evaluate model performance is to use numerical criteria, but then the user may get lost in a jungle of potential criteria. Why is the choice so difficult? Several reasons can be put forward:

1. flows vary by several orders of magnitude that may not be equally useful for the modeller;

2. hydrological models often produce heteroscedastic errors, i.e. their variance is not independent of the flow value;
3. the range of target flows may vary significantly between evaluation periods;
4. the model may be used for different applications, which may require specific criteria.

For these reasons, a large variety of criteria have been proposed and used over the years in hydrological modelling, as shown for example by the lists of criteria given by the ASCE (1993), Dawson et al. (2007), Moriasi et al. (2007) and Reusser et al. (2009). Among these criteria, some are *absolute* criteria such as the widely used root mean square error, while others are *relative* criteria (i.e. normalized) such as the Nash and Sutcliffe (1970) efficiency criterion (NSE). In the latter, model errors are compared to the errors of a reference or benchmark model (Seibert, 2001; Perrin et al., 2006). This provides a useful quantification of model performance in that it indicates to which extent the model is better (or worse) than the benchmark. It also facilitates the comparison of performance between catchments.

However, the choice of a benchmark is difficult: different benchmarks are more or less demanding (and thus the comparison more or less informative) depending on the type of hydrological regime or the type of model application. This sometimes makes it difficult to interpret the relative criteria and a bit puzzling for an inexperienced

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between graphs, all series were normalized by the maximum value in the series, thus providing values ranging between 0 and 1). Fig. 1 illustrates the significant differences between transformations and the resulting error values: indeed, the part of the hydrograph that produces most of the errors strongly depends on the transformation chosen. Hence, a detailed analysis of power law transformations is needed and will be discussed in Section 3.6.

1.4. Scope of the article

The present work intends to complement previous studies by objectively identifying which criteria are best suited for the evaluation of low-flow simulations using actually observed data. This paper has two main objectives:

- First, we wish to discuss the links between the criteria used or proposed to evaluate low-flow simulations.
- Second, we wish to better understand which part of the hydrograph most influences the various criteria used for the evaluation of low-flow simulations.

To meet these objectives, we will evaluate various criteria, mainly pertaining to the family of criteria proposed by Le Moine (2008) in the form of the Nash–Sutcliffe efficiency index. They will be compared to other criteria proposed in the literature for

low-flow evaluation on a large set of catchments showing various low-flow characteristics. We will restrict this analysis to (i) the case of nondimensional criteria (i.e. relative criteria in which the model error is normalized), and (ii) deterministic low-flow simulations. Although several authors recently questioned the relevance of deterministic predictions in an information theory perspective (Weijs et al., 2010), the criteria for the evaluation of probabilistic low-flow predictions are not within the scope of this study (see e.g. Laio and Tamea (2007), Bartholmes et al. (2009), Boucher et al. (2009), Ramos et al. (2010) or Randrianasolo et al. (2010) for discussions on this issue). Last, we will not discuss the issue of model calibration (i.e. the choice of objective function), but concentrate on the evaluation of simulations only.

The next section presents the data set and methodology used in this study. Then the results are detailed and discussed before the concluding remarks.

2. Data set, models and evaluation methodology

2.1. Data set

To carry out the evaluation, a set of 940 catchments throughout France was used to test the models in various hydro-meteorological conditions, including oceanic, Mediterranean and continental

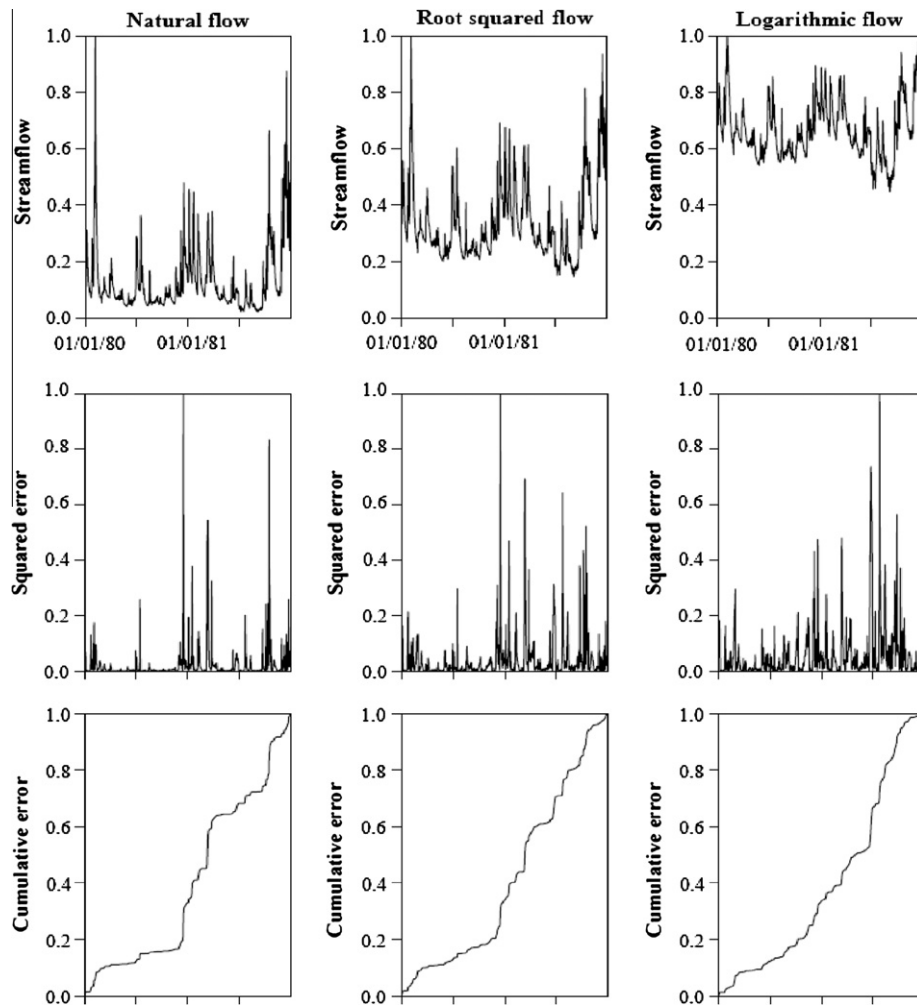


Fig. 1. Illustration of changes in the series of flows, model errors and cumulated squared errors using root square and logarithm transformations for the Moselle River at Custines (Hydro code A7010610; catchment area 6830 km²; period: 01/01/1980–31/12/1981). All series are normalized between 0 and 1 by dividing by the maximum value over the period.

influences. This should provide more robust conclusions than testing the models on a single catchment (Andréassian et al., 2006a). Detailed information on this data set can be found in Le Moine et al. (2007) and Le Moine (2008). A few characteristics are given in Table 2. The catchments studied are all unregulated, although some of them may be influenced in the summer periods by human withdrawals.

Daily time series were used, as this time step is well adapted to the study of low flows. Continuous series of precipitation, potential evapotranspiration and streamflow were available for the 1970–2006 time period, providing good variability of meteorological conditions, with quite severe drought periods (e.g. the years 1976, 1989–1991, 2003 and 2005). Meteorological data come from the SAFRAN reanalysis of Météo-France (Quintana-Segui et al., 2008; Vidal et al., 2010). Daily potential evapotranspiration was estimated using the formulation proposed by Oudin et al. (2005) based on temperature and extra-terrestrial radiation.

To characterize the variability in low-flow characteristics in the catchment set, we calculated the ratio between Q_{90} (the flow exceeded 90% of the time) and Q_{50} (the median flow) for each catchment, to obtain a nondimensional indicator of the severity of low flows (Smakhtin, 2001). The less variable the flow regime, the higher the ratio value. As shown in Table 2 and Fig. 2, the catchments in this data set experience a large variety of low-flow conditions. Some catchments seem to have a very smoothed response with substantial low flows, while others seem much more responsive with high flow variability. A few catchments have an index value of zero, corresponding to a Q_{90} value of zero, which means that they are nonperennial catchments.

2.2. Hydrological models

The tests were carried out using two continuous lumped rainfall–runoff models: the four-parameter GR4J model (Perrin et al., 2003) and a six-parameter version of the MORDOR model (Garçon, 1999; Mathevet, 2005) called MORD hereafter to avoid confusion with the original model. These models are quite widely used in France for operational purposes. Schematic diagrams of model structures are shown in Fig. 3. For detailed descriptions of the models, please refer to Andréassian et al. (2006b). Here, models were used in the same conditions, i.e. they were fed exactly with the same rainfall and potential evapotranspiration inputs.

The lumped modelling approach adopted here is often thought unsuitable for quite large catchments like some of those used here (catchment area goes up to almost 10,000 km² in our catchment set). However, contrary to this preconceived idea, we noticed that the median Nash–Sutcliffe efficiency tends to increase with catchment size for the tested models on our catchment set (not shown here). This can be related to the results presented by Merz et al. (2009) who applied a hydrological model (with semi-distributed inputs but lumped parameters) to 269 Austrian catchments ranging in size from 10 to 130,000 km². They concluded that model efficiency tends to increase over the scale range of 10–10,000 km².

Note that the two models simulate the low flows in a quite different manner: in the GR4J model, low flows are generated by the percolation from the soil moisture store (S), outflow from the

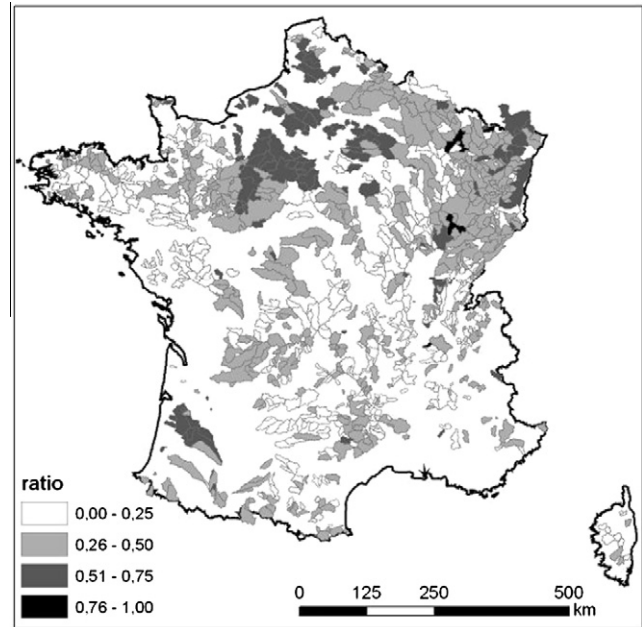


Fig. 2. Spatial distribution of the Q_{90}/Q_{50} index values over the set of 940 French catchments.

routing store (R) and the groundwater exchange function (F) (which simulates possible interactions with deep aquifers or surrounding catchments). However, in MORD, low flows are produced by two routing stores (L and N) (Fig. 3).

The models' parameters were calibrated using a two-step search procedure (a prior gross inspection of the parameter space to identify the most likely zone of convergence, followed by a local search algorithm). This approach proved efficient for these parsimonious models compared to more complex search algorithms (see Edijatno et al., 1999; Mathevet, 2005).

As an objective function for calibration, we used the root mean squared error calculated on root square flows, which was found by Oudin et al. (2006) to be a good compromise for an all-purpose model (not giving too much emphasis to low or high flows). Discussing the impact of the choice of the objective function on the quality of low-flow simulations is not within the scope of this article, and we will only examine the relevance of different criteria for model evaluation in validation mode.

2.3. Testing scheme

This section describes the method used to evaluate criteria. The models were run following the split sample testing scheme proposed by Klemes (1986), i.e. splitting the whole available record into two independent periods (P1 and P2) of equal length, calibrating the model on P1 and validating it on P2 and then exchanging the roles of P1 and P2. Hence the models could be evaluated in validation over the whole data record. The first year of each test period was used for model warm-up. In the following, the results shown correspond only to validation tests. To summarize the

Table 2
Characteristics of the 940 test catchments.

Catchment characteristics	Exceedance percentiles						
	1.00	0.95	0.75	0.50	0.25	0.05	0.00
Catchment area (km ²)	5.3	2075	391	162	78	28	9423
Mean annual precipitation (mm/y)	621	1548	1152	964	828	714	2143
Mean annual potential evapotranspiration (mm/y)	337	750	685	645	615	533	814
Mean annual runoff (mm/y)	50	1201	561	366	244	141	3689
Q_{90}/Q_{50} (–)	0.00	0.64	0.39	0.28	0.18	0.06	0.94

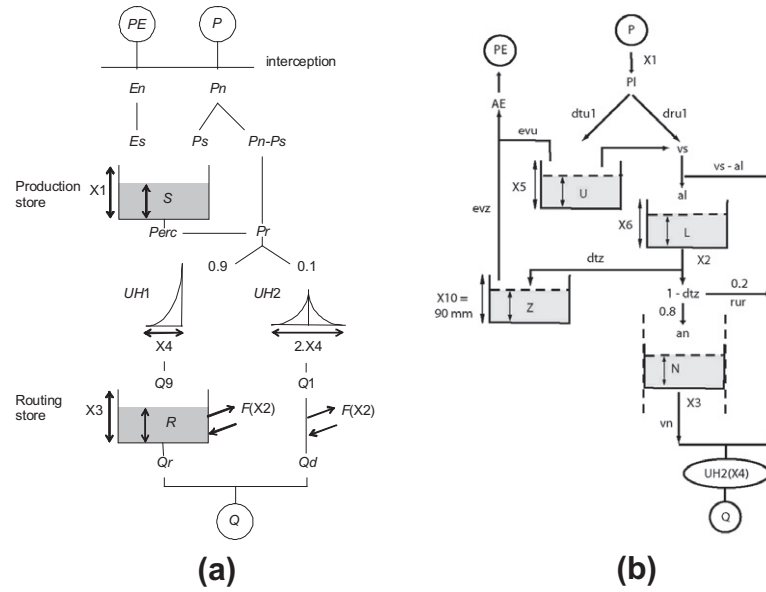


Fig. 3. Schematic diagrams of the structures of (a) the GR4J model and (b) the MORD model (PE: potential evapotranspiration; P: rainfall; Q: streamflow; X: parameters; other letters are internal model variables).

results on the whole set of catchments, we mainly used box plots based on the 940×2 results obtained in validation.

2.4. Criteria analysed

The analysis was carried out on the criteria detailed in Table 3:

- The first five criteria represent different Nash–Sutcliffe efficiencies calculated on flows (NSE_Q), root squared flows ($NSE_{\sqrt{Q}}$), logarithmic transformed flows ($NSE_{\ln Q}$), inverse flows ($NSE_{1/Q}$) and relative flows (NSE_{rel}). $NSE_{1/Q}$ is not commonly used in the literature and was suggested by Le Moine (2008). The comparison of these criteria will provide information on the effect of prior flow transformation.
- NSE_{uT} is the NSE calculated under a threshold taken equal to the 0.9 percentile of the observed flow duration curve of the test period (other thresholds could obviously be considered). Compared to the previous criteria, NSE_{uT} is calculated only on one-tenth of the time steps corresponding to the lowest observed flows and will provide information on the use of a threshold.
- IA_{rel} is the relative index of agreement proposed by Krause et al. (2005). Here, the normalization of the error is different from the one made in the NSE .
- B_{uT} is the bias under the same threshold as for NSE_{uT} (0.9 flow percentile). Contrary to the previous criteria that are all based on the squared error, B_{uT} is based on the cumulative error and will indicate the error of water balance in low-flow conditions.
- Last, $RLFD_{uT}$ is the flow deficit under the low-flow threshold (0.9 flow percentile). The formulation of $RLFD_{uT}$ is similar to that of B_{uT} and indicates the error of the water deficit in low-flow conditions based on the threshold.

As these criteria have no lower bounds, they may take strongly negative values in case of model failure. These individual values can severely impact the calculation of statistical moments of the set of criteria (e.g. the mean value). To avoid this bias, we used the mathematical transformation proposed by Mathevet et al. (2006), in which the range of variation of the transformed criterion is $]-1; 1]$ instead of $]-\infty; 1]$. The transformation is given by:

$$C^* = \frac{C}{2 - C} \quad (2)$$

where C^* is the transformed criterion and C is the initial one. In addition to solving the problem of extreme negative values, the 0 and 1 values remain the same: $C^* = 0$ for $C = 0$ and $C^* = 1$ for $C = 1$, hence keeping the same interpretation relative to the benchmark. Note that for $C > 0$, C^* will be lower than C and the reverse for $C < 0$.

In the next sections, we will use the notations NSE_Q^* , $NSE_{\sqrt{Q}}^*$, $NSE_{\ln Q}^*$, $NSE_{1/Q}^*$, $RLFD_{uT}^*$ for the bounded versions of the criteria listed in Table 3.

Note that for some criteria (e.g. $NSE_{\ln Q}^*$ and $NSE_{1/Q}^*$), it is not mathematically possible to compute the transformed flow values when simulated or observed flows equal zero. However, these time steps should not be ignored during model evaluation, or else it could result in an overly optimistic assessment of model performance. To avoid this, one can add a small quantity to the flow value before computing its transformation. It can be, for example, a small fraction of the mean observed flow over the test period. In the following, ε will be set at one hundredth of the mean flow, but the choice of this value will be further discussed in Section 3.5. Note that the same comment applies to all criteria derived from Eq. (1) with negative λ values.

2.5. Approach for criterion analysis

The main objective of this study was to determine which criteria were actually the best adapted to the evaluation of low flows, by an objective analysis on a large set of catchments. To this aim, we identified the part of the hydrograph bearing most of the total model error. The procedure adopted is as follows:

1. Rank daily flow values over the study period by decreasing order. For each of them, the corresponding model squared error term SE can be computed.
2. Compute the empirical cumulative frequency, k/N , with k the rank of the flow value and N is the total number of time steps.
3. Compute the weighted empirical cumulative frequency by using the values of errors SE as weights w . The weighted cumulative frequency for the k th flow is therefore given by the ratio between the accumulated error for the k smallest flows and the total error (SSE) over the N time steps:

Table 3
Selection of evaluation criteria and their corresponding formulation and specific values (Q_i and $\hat{Q}_i$ are the observed and simulated flows, respectively, n the total number of time steps, T a low-flow threshold (here $T = Q_{90}$), $\bar{Q}$ the mean of Q , and ε a small constant).

Criterion	Usual name	References	Mathematical formulation	Range of values	Value of perfect agreement	Target variable	Benchmark model
NSE_Q	NSE calculated on flows	Nash and Sutcliffe (1970)	$1 - \frac{\sum_{i=1}^n (Q_i - \hat{Q}_i)^2}{\sum_{i=1}^n (Q_i - \bar{Q})^2}$	$]-\infty, 1]$	1	Q_i	$\bar{Q}_i$
$NSE_{\sqrt{Q}Q}$	NSE calculated on root squared transformed flows	Oudin et al. (2006)	$1 - \frac{\sum_{i=1}^n (\sqrt{Q_i} - \sqrt{\hat{Q}_i})^2}{\sum_{i=1}^n (\sqrt{Q_i} - \sqrt{\bar{Q}_i})^2}$	$]-\infty, 1]$	1	$\sqrt{Q_i}$	$\sqrt{\bar{Q}_i}$
$NSE_{\ln Q}$	NSE calculated on log transformed flows	Oudin et al. (2006)	$1 - \frac{\sum_{i=1}^n (\ln(Q_i + \varepsilon) - \ln(\hat{Q}_i + \varepsilon))^2}{\sum_{i=1}^n (\ln(Q_i + \varepsilon) - \ln(\bar{Q}_i + \varepsilon))^2}$	$]-\infty, 1]$	1	$\ln(Q_i + \varepsilon)$	$\ln(\bar{Q}_i + \varepsilon)$
$NSE_{1/Q}$	NSE calculated on inverse transformed flows	Le Moine (2008)	$1 - \frac{\sum_{i=1}^n \left(\frac{1}{Q_i + \varepsilon} - \frac{1}{\hat{Q}_i + \varepsilon} \right)^2}{\sum_{i=1}^n \left(\frac{1}{Q_i + \varepsilon} - \frac{1}{\bar{Q}_i + \varepsilon} \right)^2}$	$]-\infty, 1]$	1	$\frac{1}{Q_i + \varepsilon}$	$\frac{1}{\bar{Q}_i + \varepsilon}$
NSE_{rQ}	NSE calculated on relative flows	Krause et al. (2005)	$1 - \frac{\sum_{i=1}^n ((Q_i - \hat{Q}_i) / (Q_i + \varepsilon))^2}{\sum_{i=1}^n ((Q_i - \bar{Q}_i) / (\bar{Q}_i + \varepsilon))^2}$	$]-\infty, 1]$	1	$\frac{Q_i}{Q_i + \varepsilon}$	1
NSE_{uT}	NSE calculated under low-flow threshold		$1 - \frac{\sum_{i \in [1, n] Q_i < T} (Q_i - \hat{Q}_i)^2}{\sum_{i \in [1, n] Q_i < T} (Q_i - \bar{Q}_i)^2}$	$]-\infty, 1]$	1	$Q_i < Q_{90}$	$\bar{Q}_i (Q_i < Q_{90})$
IA_{rel}	Relative index of agreement	Krause et al. (2005)	$1 - \frac{\sum_{i=1}^n ((Q_i - \hat{Q}_i) / Q_i)^2}{\sum_{i=1}^n ((Q_i - \bar{Q}_i) / (\bar{Q}_i / Q_i))^2}$	$]-\infty, 1]$	1	$\frac{Q_i}{Q_i}$	$\bar{Q}_i$
B_{uT}	Bias under low-flow threshold		$1 - \frac{\sum_{i \in [1, n] Q_i < T} \hat{Q}_i}{\sum_{i \in [1, n] Q_i < T} Q_i}$	$]-\infty, 1]$	0	$Q_i < Q_{90}$	–
$RLFD_{uT}$	Ratio of Low-flow deficit		$1 - \frac{\sum_{i \in [1, n] Q_i < T} \max(0, T - \hat{Q}_i)}{\sum_{i \in [1, n] Q_i < T} (T - Q_i)}$	$]-\infty, 1]$	0	$T < Q_i$	–

$$F_k = \frac{\sum_{i=1}^k w_i}{\sum_{i=1}^N w_i} = \frac{\sum_{i=1}^k SE_i}{\sum_{i=1}^N SE_i} = \frac{\sum_{i=1}^k SE_i}{SSE} \quad (3)$$

- Summarize the weighted cumulative distribution as a box plot showing the flow percentiles of exceedance: 0.05, 0.25, 0.50, 0.75 and 0.95.
- Transform back these flow percentiles into percentiles of the flow duration curve. The range between the 0.05 and 0.95 percentiles gives a 90% confidence interval of the range of observed flows contributing the most to the total model error.

An example of weighted and non weighted distributions and the results of steps 4 and 5 above are illustrated in Fig. 4. For the same example, Fig. 5 illustrates flow values in the hydrograph that contributes most of the model error. In this figure, the darker the observation, the larger the contribution of this time steps to the total model error.

This was done for each criterion and for each catchment. The resulting box plots can be compared between catchments. We did not analyse the results for each catchment individually, but for the whole catchment set to obtain more general conclusions. To this aim, we will draw a “master” box plot, in which each percentile shown will correspond to the median value of the same percentiles of all the box plots computed over the catchment set. An illustration of the derivation of the master box plot is presented in Fig. 6.

3. Results and discussion

This section first presents the distributions of the efficiency criteria over the 940 catchments and then analyses their interdependency and their behaviour in terms of dependence to low-flow values. This leads to a selection of criteria for the evaluation of low-flow simulation.

3.1. Overall efficiency distribution

Fig. 7 shows the box plots of the distributions of efficiency values for the entire catchment set (i.e. the 940×2 values obtained in

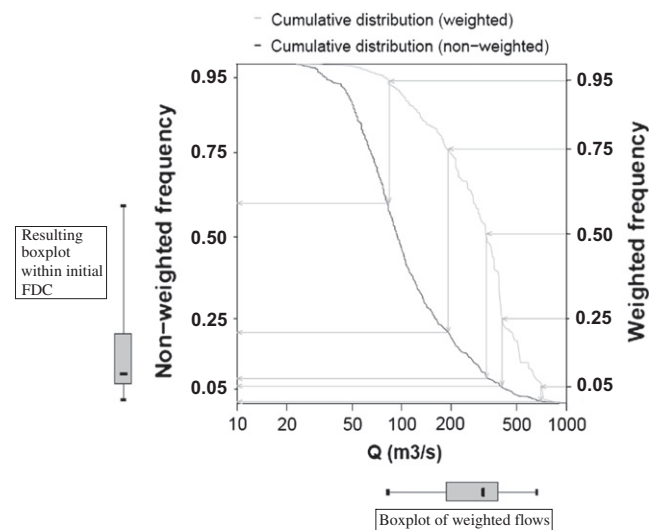


Fig. 4. Example of derivation of box plots from the distribution of weighted flows (here weights are the squared errors) to the flow duration curve (distribution of non weighted flows). The same example as in Fig. 1 is used here (period 01/01/1980–31/12/1981).

validation) and for the nine criteria. The two models show quite similar behaviours.

The NSE_{uT}^* (NSE calculated under 0.9 exceedance flow percentile) exhibits very different behaviour from the other criteria. Its values mostly lie between -1 and 0 . This indicates that on most catchments, the two models used are not better than the benchmark that provides the mean observed value under the threshold as the simulation. Although this result is a bit disappointing, it should be remembered that the variability of flow values in these conditions is limited, so that the benchmark is already a model that is difficult to equal (even if a constant flow is not perfect in terms of dynamics, it is already quite good). Besides, the benchmark is by construction perfect in terms of water balance, which means that no model can be better (see e.g. Perrin et al., 2006). Therefore, although they are clear in terms of the level of performance provided by the models, these mostly negative values make the crite-

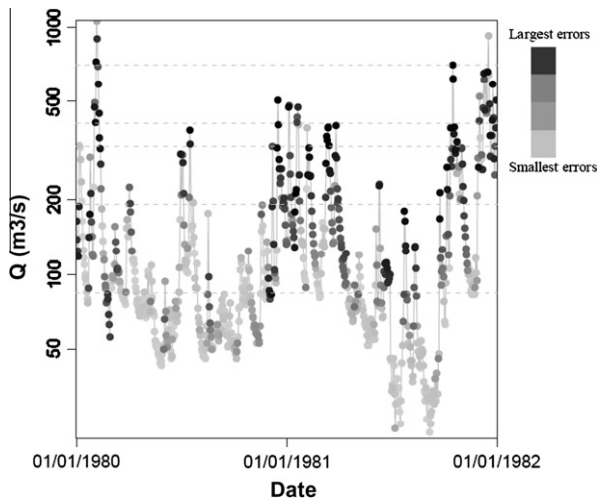


Fig. 5. Illustration of flow values that contribute the most of the total model error. The error considered here is the square error calculated on flows. The darker the dots, the larger the contribution to the total error. The same example hydrograph as in Fig. 1 is used here. Horizontal dashed lines correspond to the percentiles of the box plot shown in Fig. 4.

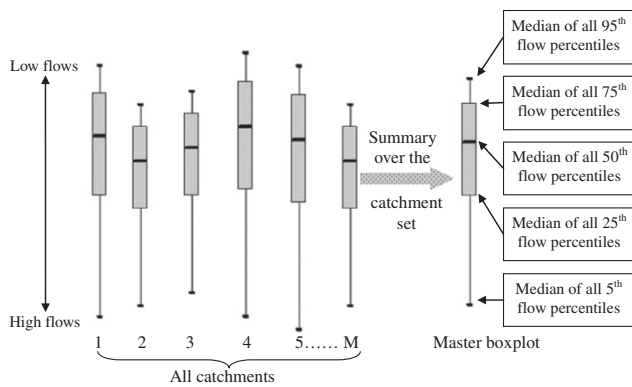


Fig. 6. Schematic representation of the derivation of master box plot which summarizes the behaviour of a given criterion over the entire catchment set (M stands for the total number of catchments).

ria difficult to interpret: the benchmark is probably too demanding in this case. The distributions of the other efficiency criteria show a much larger share of positive values (Fig. 7).

This variety of efficiency ratings obtained with the different criteria comes clearly from the differences in criteria formulations that put more or less emphasis on different parts of the hydrograph. This indicates that the tendency of modellers to declare that a model is “bad” or “good” based on the interpretation of relative efficiency criteria is often dangerous: it would be more rigorous to always refer to the benchmark used to say to what extent the model is “worse” or “better” than this benchmark.

3.2. Criteria interdependency

To analyse the possible interdependency between criteria, we drew scatter plots of pairs of criteria over the catchment set. The scatter diagrams for the possible combinations of criteria are presented in Tables 4 and 5 for the GR4J and MORD models, respectively (note that patterns in scatter diagrams were similar for the two models tested, which indicates that they are not model-specific). The scatter plots of NSE_Q^* vs. NSE_{sqrQ}^* , NSE_{sqrQ}^* vs. NSE_{lnQ}^* , and NSE_{lnQ}^* vs. IA_{rel}^* illustrate that there is a significant relation between these pairs of criteria. It means that they provide similar information on

model efficiency. The relation between NSE_{rQ}^* and IA_{rel}^* could be expected since these criteria are based on the same model error and differ only in their normalization. The larger scatter between NSE_{lnQ}^* and NSE_Q^* shows that the log transformation substantially changes the information contained in the NSE_Q criterion. This is even more obvious when looking at the absence of relationships between NSE_Q^* and NSE_{iQ}^* and NSE_{rQ}^* on the peak values. For the remaining pairs of criteria, the relationships are generally weak, indicating that they probably provide complementary information.

3.3. Analysis of model error terms

As explained in Section 2.5, we built master box plots over the entire set of catchments for the nine efficiency criteria to try to identify from which part of the flow duration curve most of the model error comes on average. The results are shown in Fig. 8. Here again, the results show similar patterns for the two simulation models, indicating that they are not highly model-dependent.

As could be expected for NSE_Q^* given the past analyses on this criterion, 90% of the model error accumulates below the 20th percentile of the FDC. This confirms the bias of NSE_Q^* towards high flows and strongly supports the suitability of NSE_Q^* for the evaluation of model performance on flood events.

The box plots for the NSE_{sqrQ}^* , NSE_{lnQ}^* , and NSE_{iQ}^* criteria show that the transformations with decreasing power (root square, logarithm, inverse) progressively put more emphasis on low flows. The inverse transformation seems the most appropriate if one wishes to concentrate on the very low flows, i.e. roughly the 20% lowest flows in the hydrograph. The box plots for NSE_{rQ}^* and IA_{rel}^* are similar. Ironically, NSE_{lnQ}^* did not show clear links with NSE_{rQ}^* and IA_{rel}^* , but these criteria seem to concentrate on similar parts of the hydrograph, with a stronger emphasis on low flows. NSE_{uT}^* , B_{uT}^* and $RLFD_{uT}^*$ stress the 10% lowest flows by construction.

3.4. Which criterion should be used for low-flow evaluation?

Our results tend to confirm previous comments made in the literature:

- NSE_Q^* should only be regarded as a criterion to evaluate the high-flow simulation efficiency and is of no use for the evaluation of low-flow simulations.
- NSE_{sqrQ}^* provides more balanced information as the errors are more equally distributed on high- and low-flow parts, in agreement with the study by Oudin et al. (2006).
- NSE_{lnQ}^* is also intermediate like NSE_{rQ}^* and IA_{rel}^* but with greater emphasis on low flows. However, the sensitivity of NSE_{lnQ}^* to high flows noted by Krause et al. (2005) is confirmed here.

The NSE criterion on inverse flows (NSE_{iQ}^*) appears to be the most useful criterion to evaluate the very low flows among the criteria calculated over the total period, because it shows no sensitivity to high flows, contrary to NSE_{lnQ}^* . This transformation is more relevant than the logarithmic transformation if one wishes to evaluate models in severe low-flow conditions. In comparison with NSE_{uT}^* , B_{uT}^* and $RLFD_{uT}^*$, it avoids the often arbitrary definition of a low-flow threshold: the most widely used low flows range from Q_{70} to Q_{99} (Smakhtin, 2001) but may depend on local conditions and regime type. In addition, it avoids obtaining the often negative values found with NSE_{uT}^* , which are due to an overly demanding benchmark model and are difficult to interpret.

Therefore, we would suggest that NSE_{iQ}^* is a good candidate (the most adequate of the criteria analysed here) to evaluate the low-flow simulation efficiency of a hydrological model. We wish to emphasize that even if at present NSE_{lnQ}^* is the most commonly preferred criterion to evaluate low flows, its value is influenced by a wider range of flows.

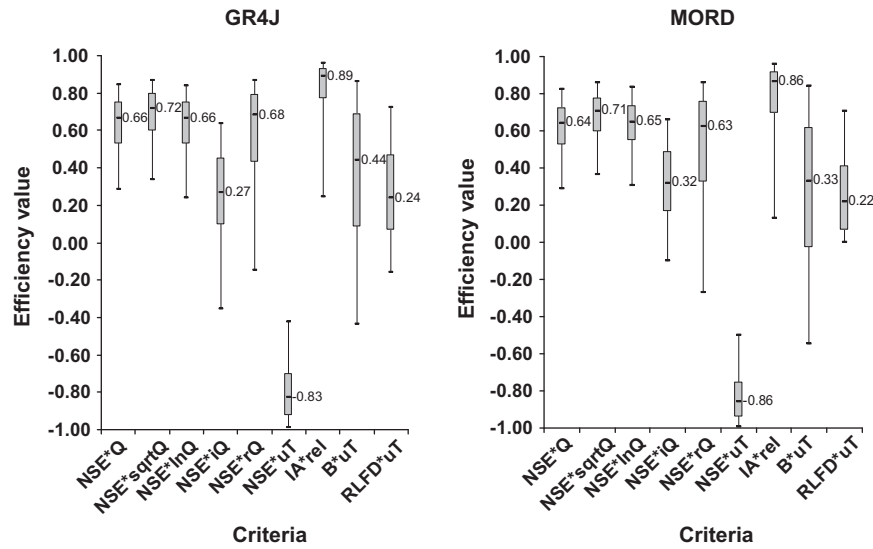
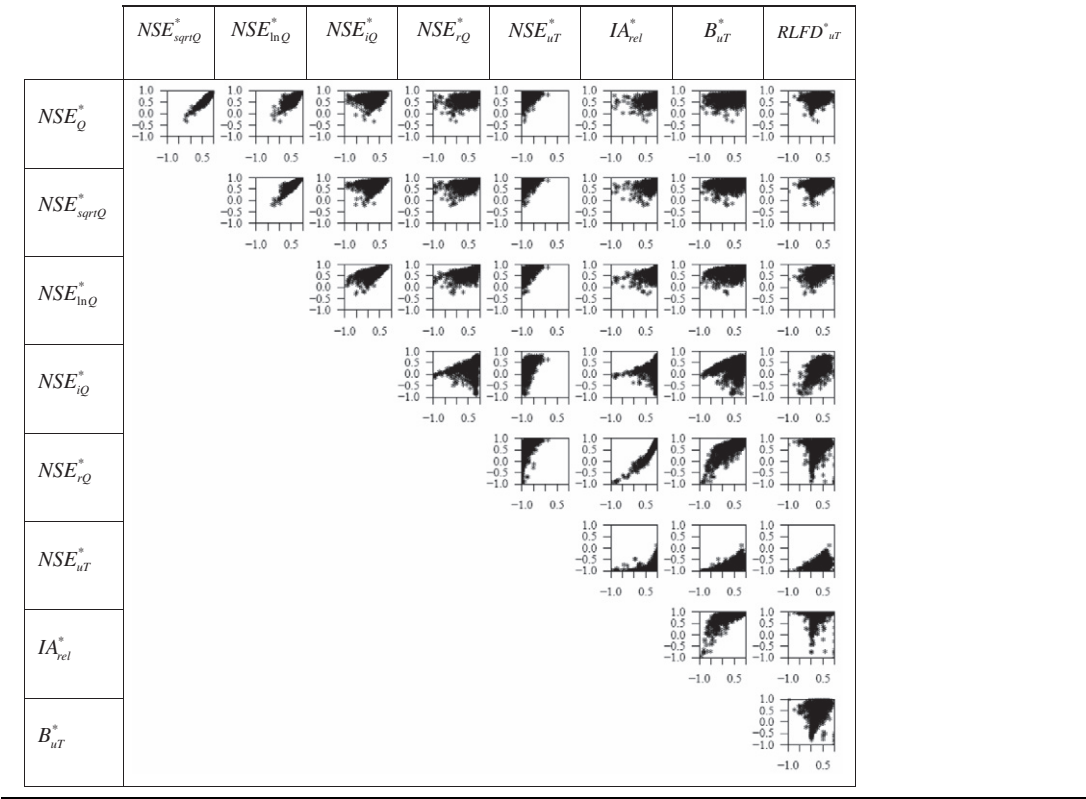


Fig. 7. Box plot of distribution of efficiency criteria obtained by the GR4J and MORD models over the entire catchment set in validation (boxes represent the 0.25 and 0.75 percentiles, with the median value inside, and the whiskers represent the 0.05 and 0.95 percentiles).

Table 4
Scatter plots of pairs of criteria on the 940-catchment set for the GR4J in validation.



3.5. What should be done with zero flows when calculating NSE_{IQ}^* ?

As mentioned in Section 2.4, the calculation of the NSE on inverse flows raises a problem when the catchment is non-perennial or ephemeral, i.e. with some days with zero flows. To deal with zero-flow values, it was necessary to add a small constant, ε , to the flow values before calculating their transforms. However, the choice of ε may have an impact on the relative level of performance

attributed to the model. As shown in Fig. 9, where values of ε ranging from one-tenth to one-hundredth of the mean flow were considered, the mean model efficiency on the catchment set varies significantly. ε should not be too large, or else it would tend to artificially enhance the relative level of model efficiency. Therefore the value of one-hundredth of the mean flow could be advised for ε , as it seems that this value corresponds to a plateau in the efficiency values, i.e. with low sensitivity of model efficiencies.

Table 5
Scatter plots of pairs of criteria on the 940-catchment set for the MORD in validation.

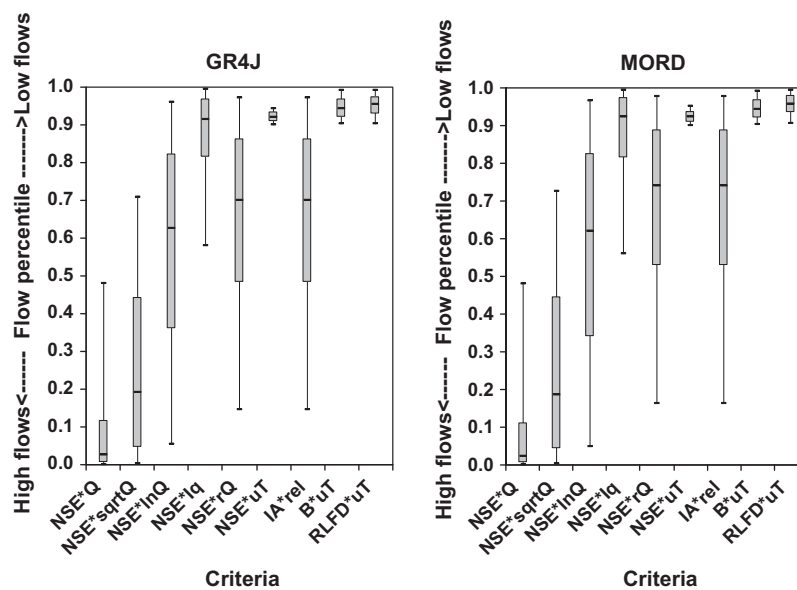
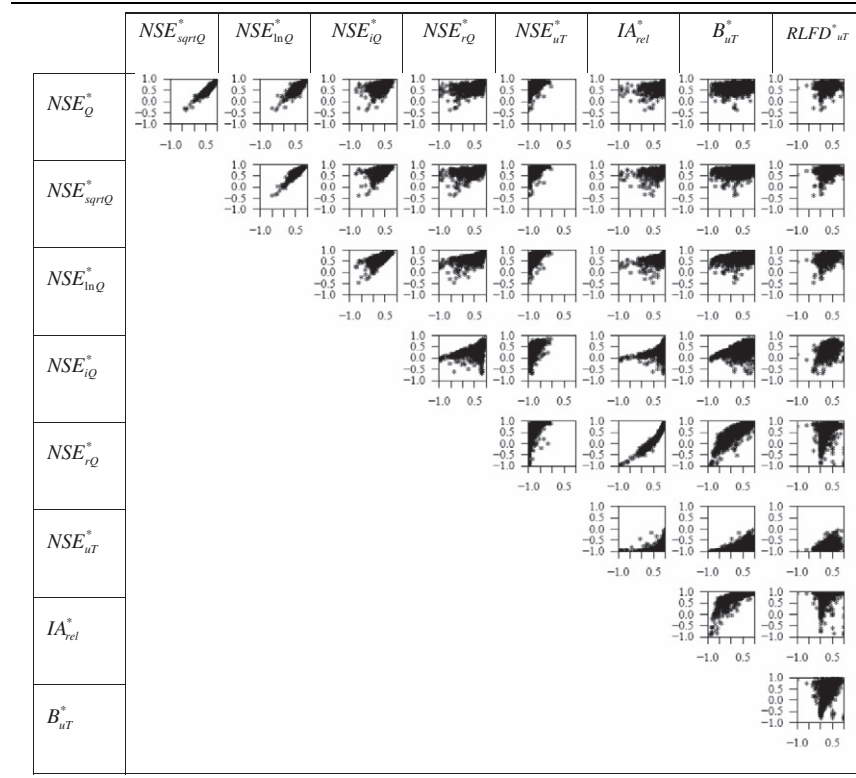


Fig. 8. Master box plot obtained over the entire catchment set for the GR4J and MORD models (each flow percentile of exceedance corresponds to the median of this percentile over all the box plots obtained on each catchment of the entire set).

3.6. Power transformation and criteria

The results in Section 3.3 clearly indicate that transformations on flows with power (λ) values lower than 0 give greater emphasis to low flows. A value of λ equal to -1 (i.e. the inverse transformation which we ultimately recommend) was shown to stress the approximately 20% of lowest flows on average on the catchment

set, as shown in Fig. 8. However, the box plots in this figure indicate that this λ value does not emphasize the quality of model fit on exactly the same parts of the hydrographs for all catchments. Indeed, it may depend on regime characteristics, flow variability or model performance on low flows. We tried to quantify the potential variability of λ values from catchment to catchment which would emphasize model errors on the same parts of the

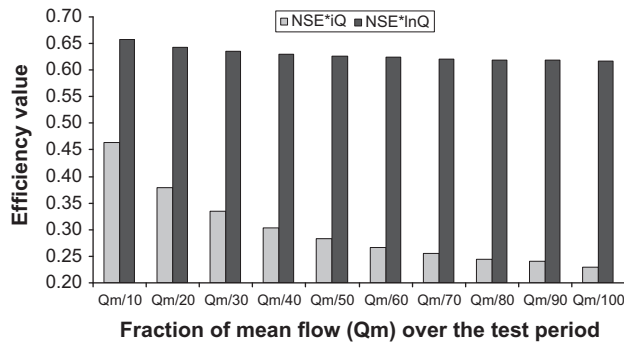


Fig. 9. Change in the mean value of NSE_{IQ}^* and NSE_{lnQ}^* obtained by the GR4J model in validation over the catchment set with different values of ε (fractions of Q_m from $Q_m/10$ to $Q_m/100$).

hydrograph. For each catchment, we selected the λ value for which 80% of the largest errors were concentrated on the 20% lowest flows at most. Fig. 10 compares the λ values obtained on the two test periods for the two models. The range of λ values obtained for the two models is similar but also quite wide, which confirms that a single λ value will not emphasize the same flow range on different catchments. The scatter on the graphs also shows a quite strong dependency of λ values on the test period. A detailed analysis of the catchments where the λ value is very unstable shows that the criterion is very sensitive to a few time steps among the lowest flows in one period where the model makes large errors. Fig. 11 compares the λ values obtained by the two models. Interestingly, there is quite good agreement between them, though the scatter is quite wide for decreasing λ values. This is probably the indication that models behave similarly in low-flow conditions, but that their behaviour on extreme low flows is different. These results indicate a dependency on the characteristics of the test period and the model used.

The dependency of λ on flow characteristics is illustrated for the GR4J model in Fig. 12. It shows the relationship between λ and the Q_{90}/Q_{50} ratio that quantifies the variability of flows. Although the scatter is large, the value of λ seems to depend on this descriptor: the smoother the catchment response (i.e. the larger Q_{90}/Q_{50}), the lower the λ value.

All these results indicate that the interpretation of the Nash–Sutcliffe criterion based on inverse transformation does require experience and that one must be cautious when comparing values obtained on different periods and/or catchments; however, this was already the case with the original Nash–Sutcliffe criterion (see the comments by Perrin et al. (2006)).

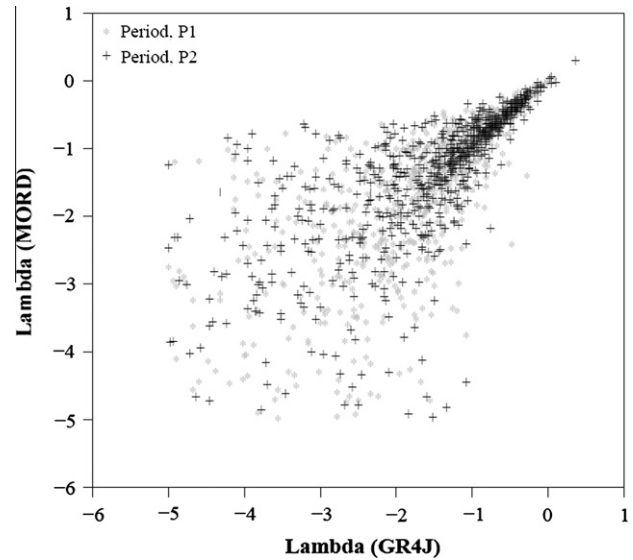


Fig. 11. Comparison of λ values obtained by the GR4J and MORD models.

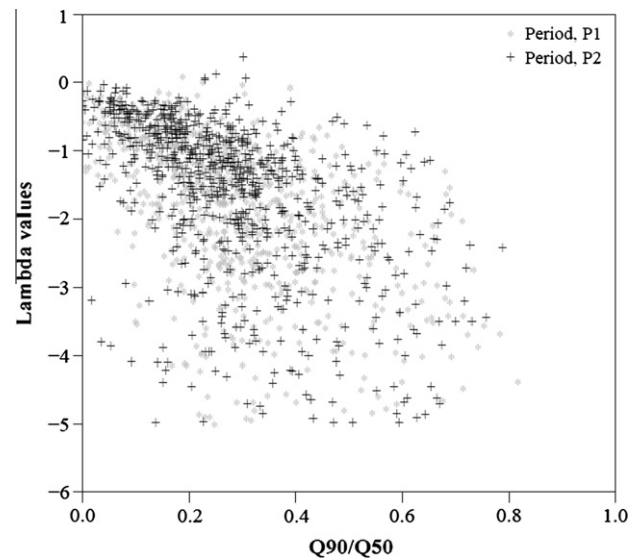


Fig. 12. Scatter plot of Q_{90}/Q_{50} vs. lambda values (λ) for the two test periods for the GR4J model.

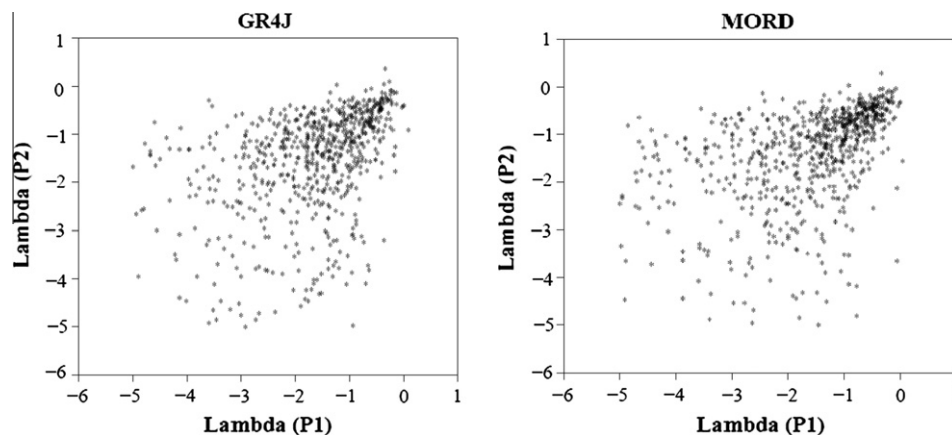


Fig. 10. Comparison of λ values obtained by GR4J (left) and MORD (right) models for the two test periods (P1 and P2).

4. Conclusions

This study aimed at identifying adequate criteria for evaluating the simulation of low flows using hydrological models. The behaviour of nine criteria was analysed on a set of 940 catchments. We used a master box plot representation to locate which part of the hydrograph contributes most of the error in these criteria, on average on the total catchment set. The detailed analysis of error distributions shows that the *NSE* calculated on inverse flows is better suited for the evaluation in very-low-flow conditions than the classically used *NSE* on logarithm flows, as it does not show sensitivity to high-flow values. It tends to focus on the 20% lowest flows on average, like the *NSE* on natural flows tends to focus on the 20% largest flows. Therefore, we recommend using this criterion calculated on inverse flow values for low-flow studies. However, it should be noted that the part of the flow duration curve that will cause most of the errors when using this transformation may differ from period to period and catchment to catchment, depending on model suitability and flow regime characteristics. This encourages to be cautious on the interpretation of this criterion when changing test conditions.

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